AGILE LAB ASSESSMENT 5

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2. Simple Static Web page.

TASK : CI and CD

- Write the code in Visual Studio and push it to your GITHUB repository.
- Modify the code and commit the updates back in GITHUB.
- Using Jenkins configure the job as pipeline and create the stages for the CI and CD process.
- Containerize the application using Dockerfile and display the stages in stage view in Jenkins.
- Create kubernetes cluster and visualize the deployment in the given node port.

b. Create some five events during Rivera.

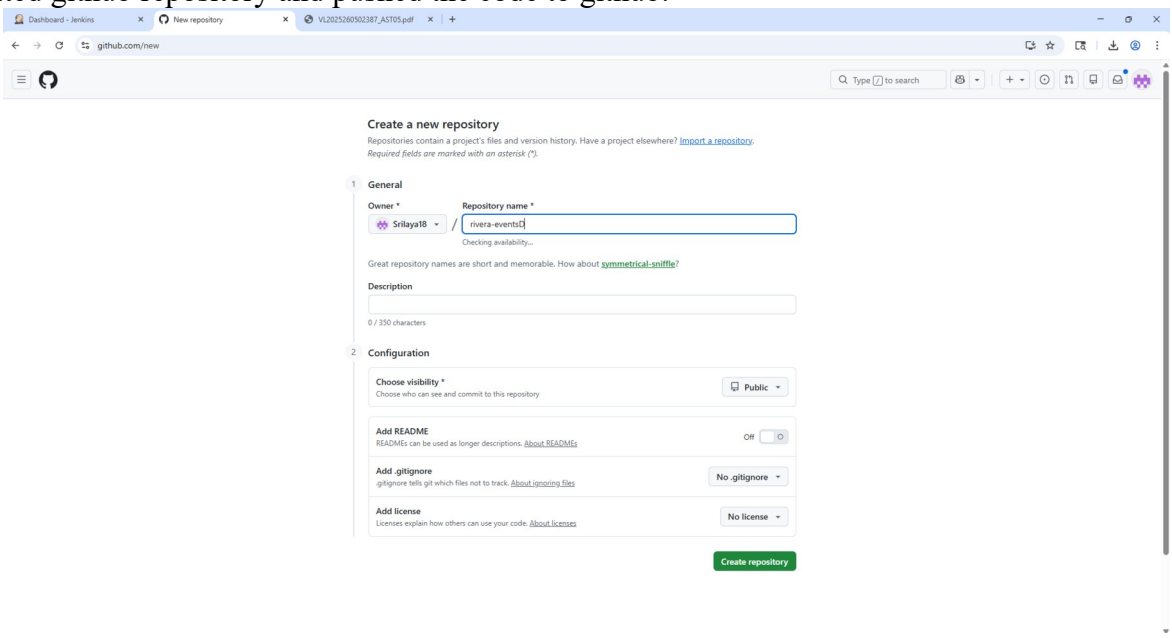
Create index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>Events During Rivera 2026</title>
    <style>
      body {
        font-family: Arial;
        text-align: center;
        background-color: #f2f2f2;
      }
      table {
        margin: auto;
        border-collapse: collapse;
        width: 70%;
      }
      th, td {
        border: 1px solid black;
        padding: 10px;
      }
      th {
        background-color: #2196F3;
        color: white;
      }
    </style>
    <body>
      <h1>Events At Rivera</h1>
      <table>
        <tr>
```

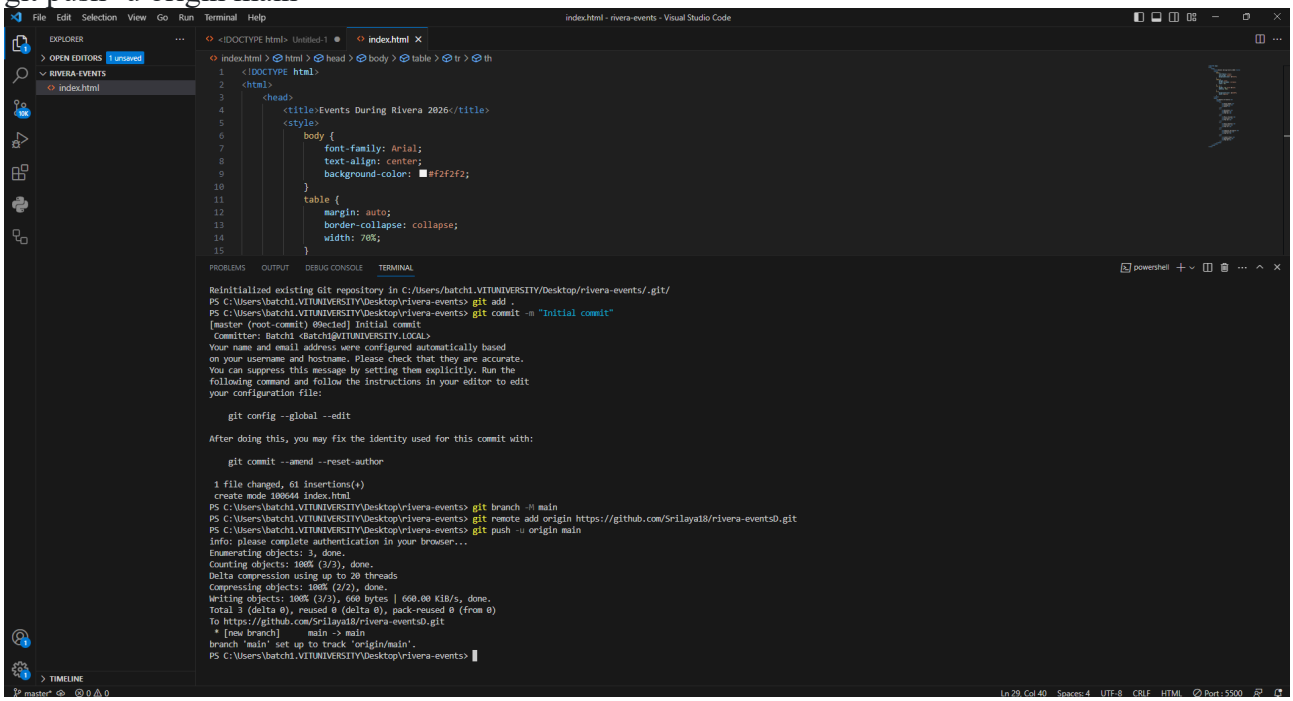
```
                <th>Event Name</th>
                <th>Category</th>
                <th>Date</th>
            </tr>
            <tr>
                <td>Marathon</td>
                <td>Health</td>
                <td>Feb 26</td>
            </tr>
            <tr>
                <td>Music Karoke</td>
                <td>Cultural</td>
                <td>Feb 26</td>
            </tr>
            <tr>
                <td>Dance Battle</td>
                <td>Cultural</td>
                <td>Feb 27</td>
            </tr>
            <tr>
                <td>Capture the Smile</td>
                <td>Creativity</td>
                <td>Feb 27</td>
            </tr>
            <tr>
                <td>Code-A-Thon</td>
                <td>Hackathon</td>
                <td>Feb 28</td>
            </tr>
        </table>
    </body>
  </head>
</html>
```

Event Name	Category	Date
Marathon	Health	Feb 26
Music Karoke	Cultural	Feb 26
Dance Battle	Cultural	Feb 27
Capture the Smile	Creativity	Feb 27
Code-A-Thon	Hackathon	Feb 28

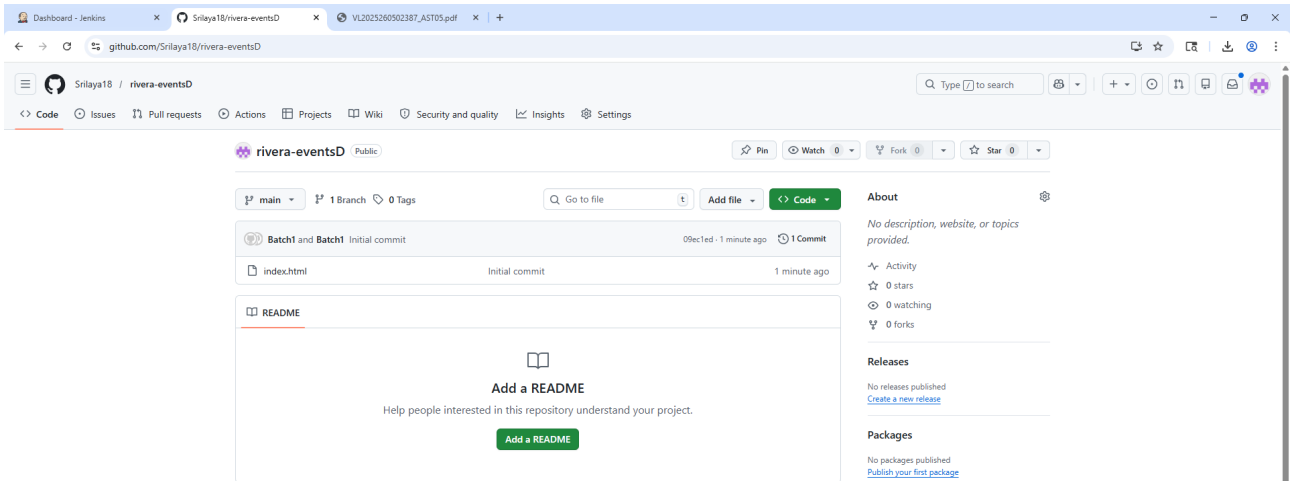
Created github repository and pushed the code to github.



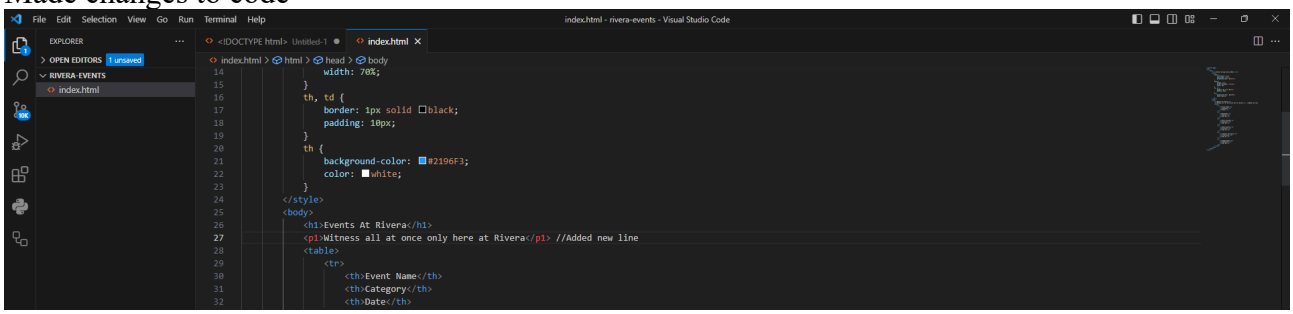
x
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/Srilaya18/rivera-eventsD.git
git push -u origin main



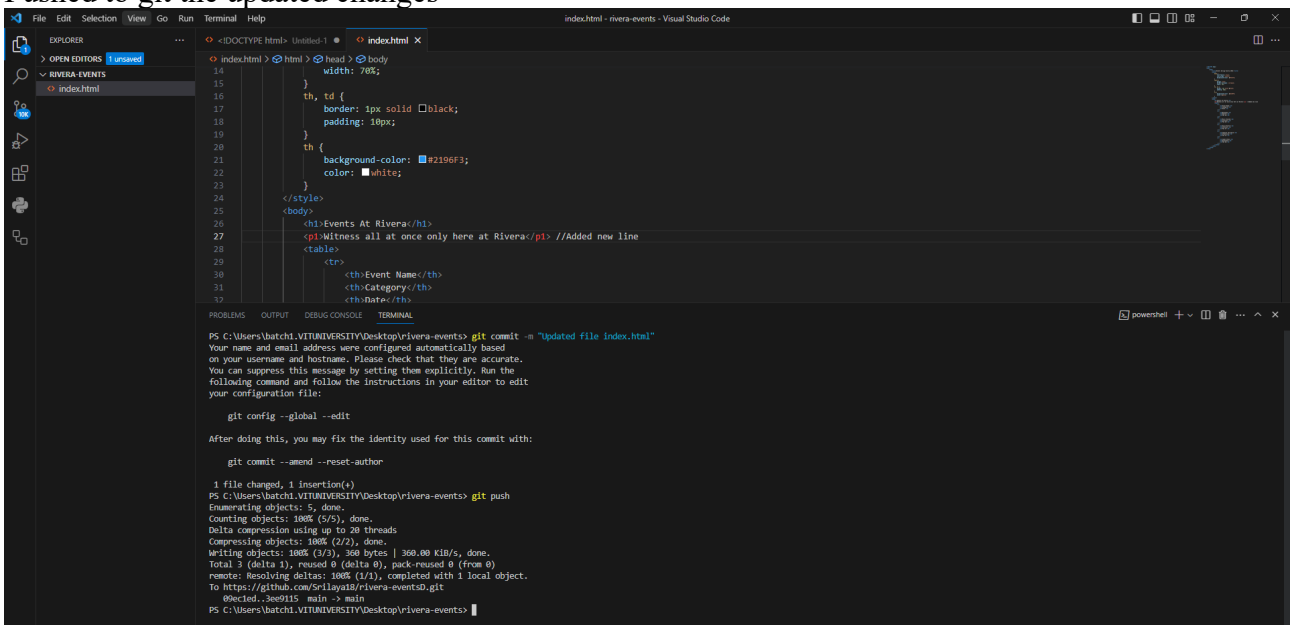
Successfully pushed to git



Made changes to code



Pushed to git the updated changes



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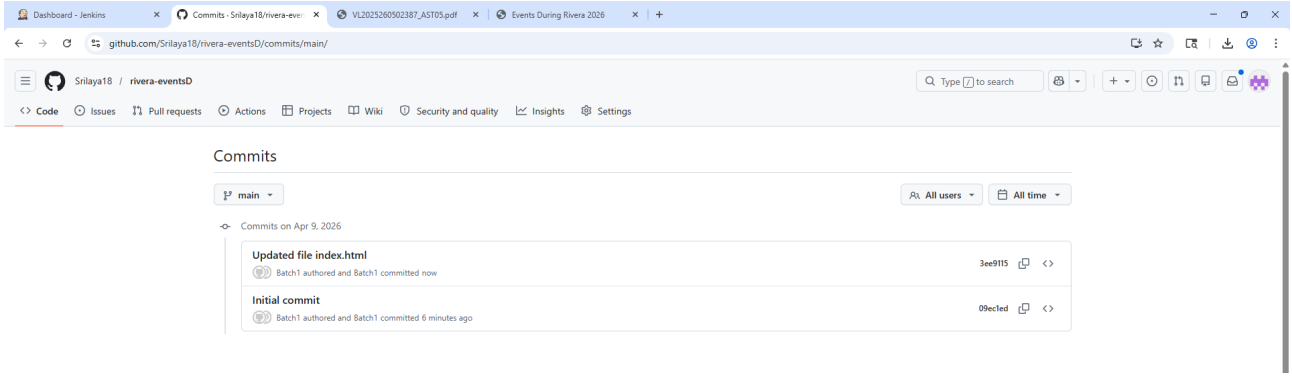


Created Dockerfile

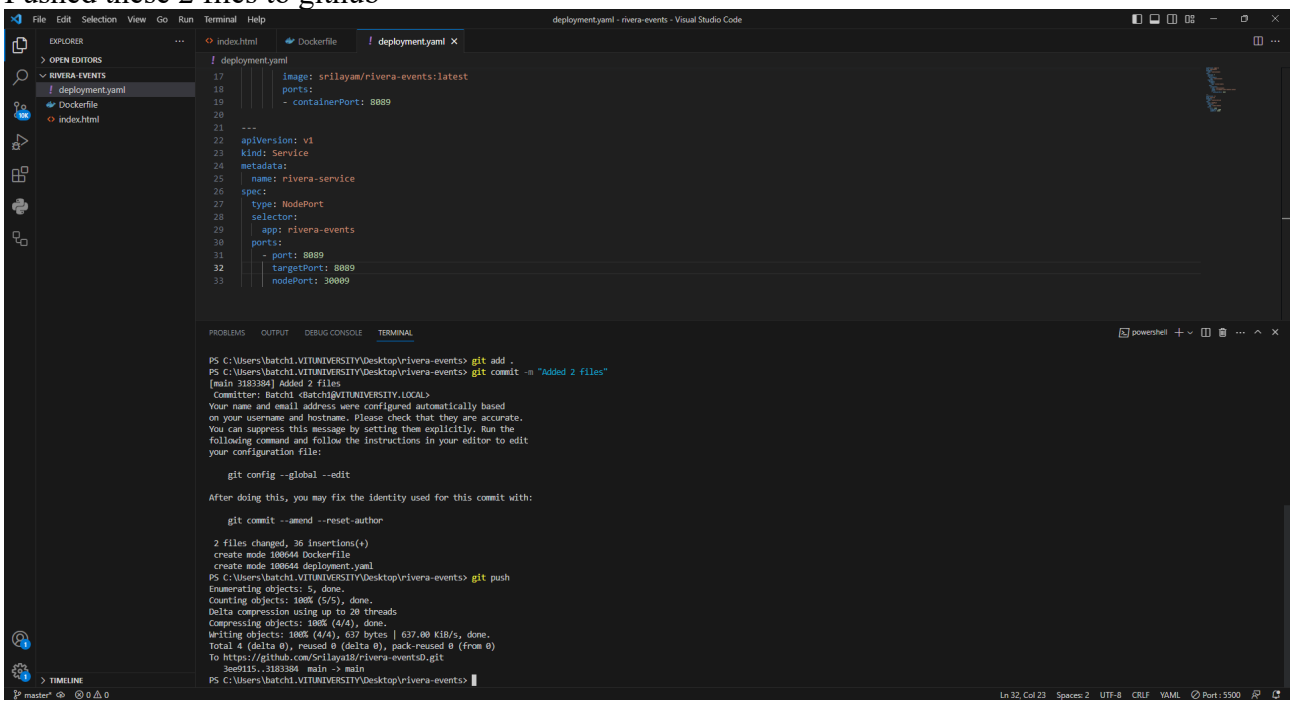
```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

Created deployment.yaml file

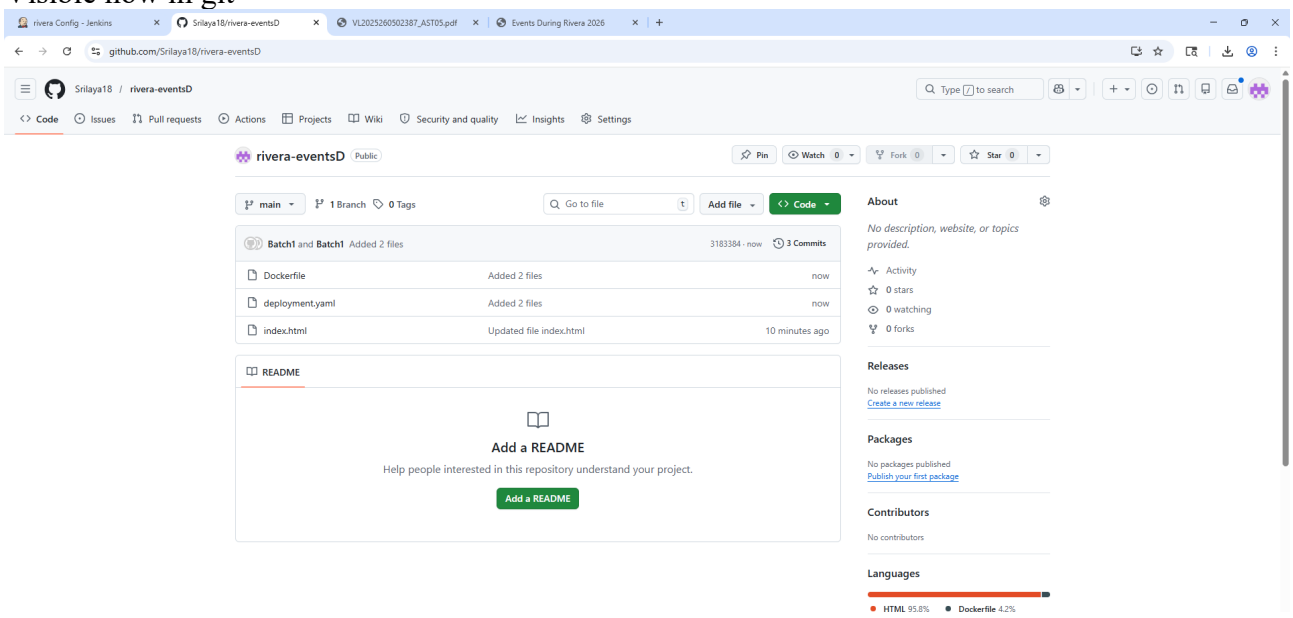
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: rivera-events
spec:
 replicas: 2
 selector:
  matchLabels:
   app: rivera-events
 template:
  metadata:
   labels:
    app: rivera-events
  spec:
   containers:
   - name: rivera-events
    image: srilayam/rivera-events:latest
    ports:
    - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
 name: rivera-service
spec:
 type: NodePort
 selector:
  app: rivera-events
 ports:
  - port: 80
   targetPort: 80
   nodePort: 30009
```

Pushed these 2 files to github

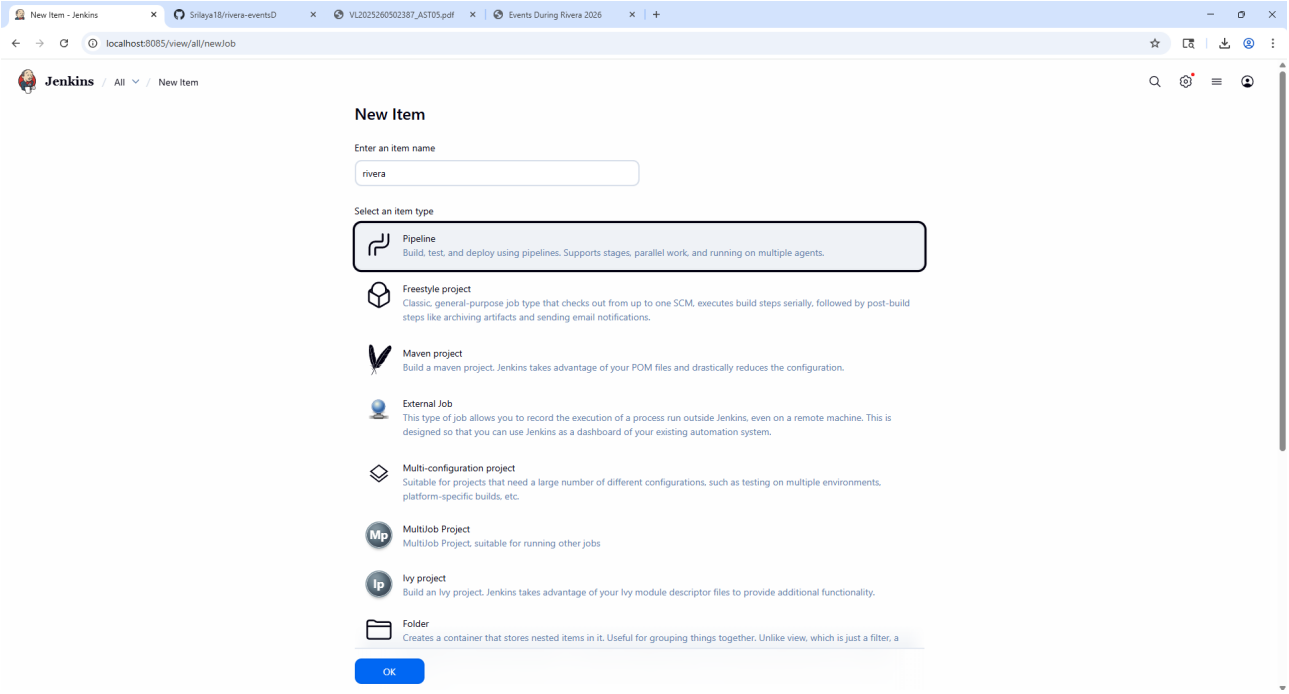


Visible now in git



s

Creating pipeline in jenkins and adding script




```
pipeline {
    agent any
    environment {
        DOCKER_IMAGE = "srilayam/rivera-events:latest"
    }
    stages {
        stage('Clone Code') {
            steps {
                git branch: 'main',
                url: 'https://github.com/Srilaya18/rivera-eventsD'
            }
        }
        stage('Build Docker Image') {
            steps {
                bat 'docker build -t %DOCKER_IMAGE% .'
            }
        }
        stage('Push Image') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'dockerhub-creds',
                    usernameVariable: 'USER',
                    passwordVariable: 'PASS'
                )]) {
                    bat 'docker login -u %USER% -p %PASS%'
                    bat 'docker push %DOCKER_IMAGE%'
                }
            }
        }
        stage('Deploy to Kubernetes') {
            steps {
                bat 'kubectl apply -f deployment.yaml --validate=false || exit 0'
            }
```

```
        }
    }
    post {
        success {
            echo 'CI/CD Pipeline executed successfully!'
        }
        failure {
            echo 'Pipeline failed. Please check logs.'
        }
    }
}
```

Deployed in jenkins

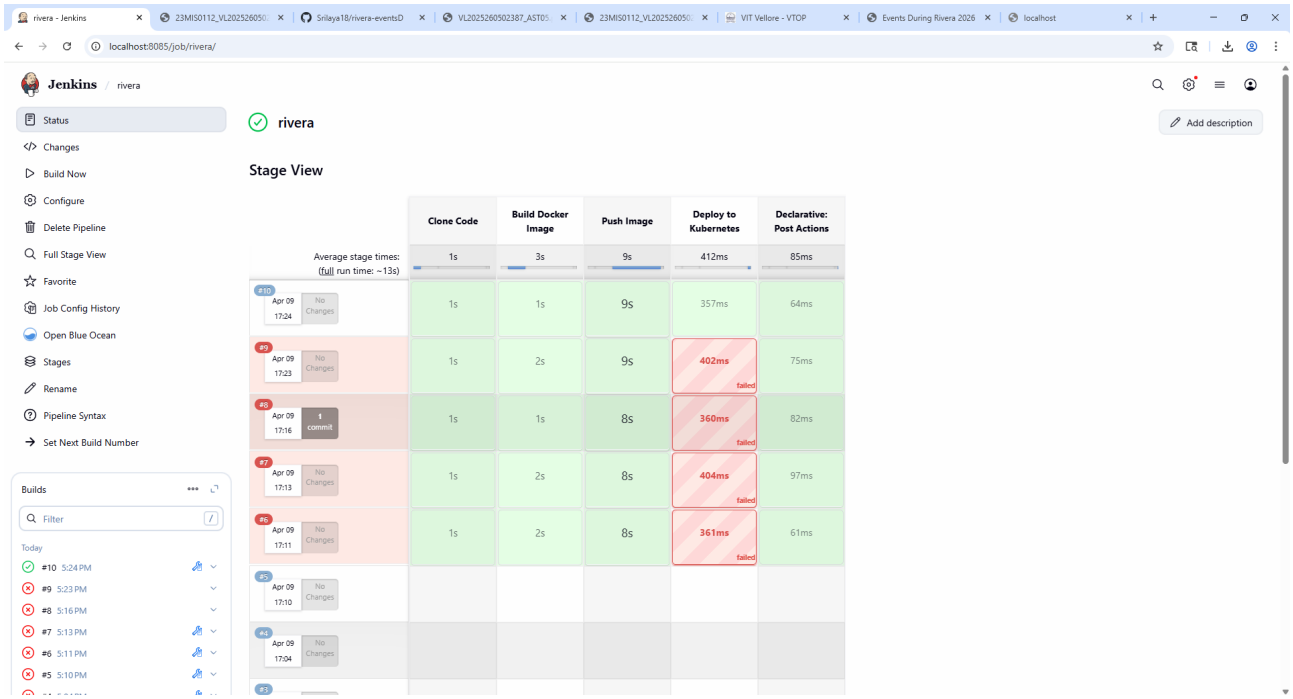
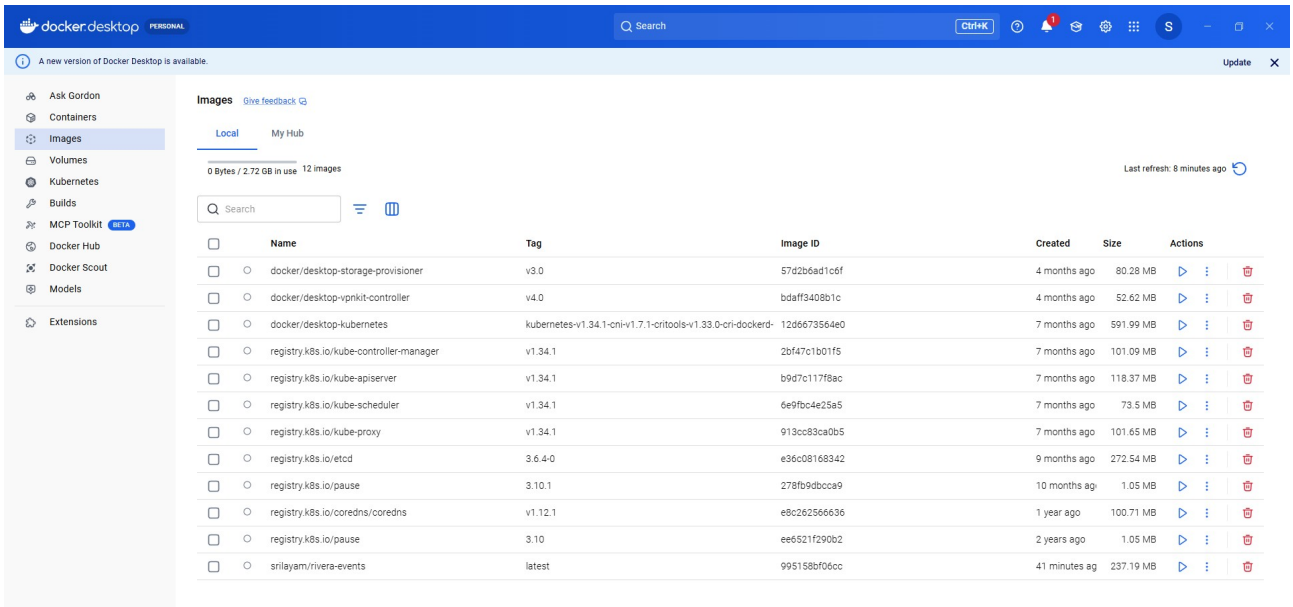
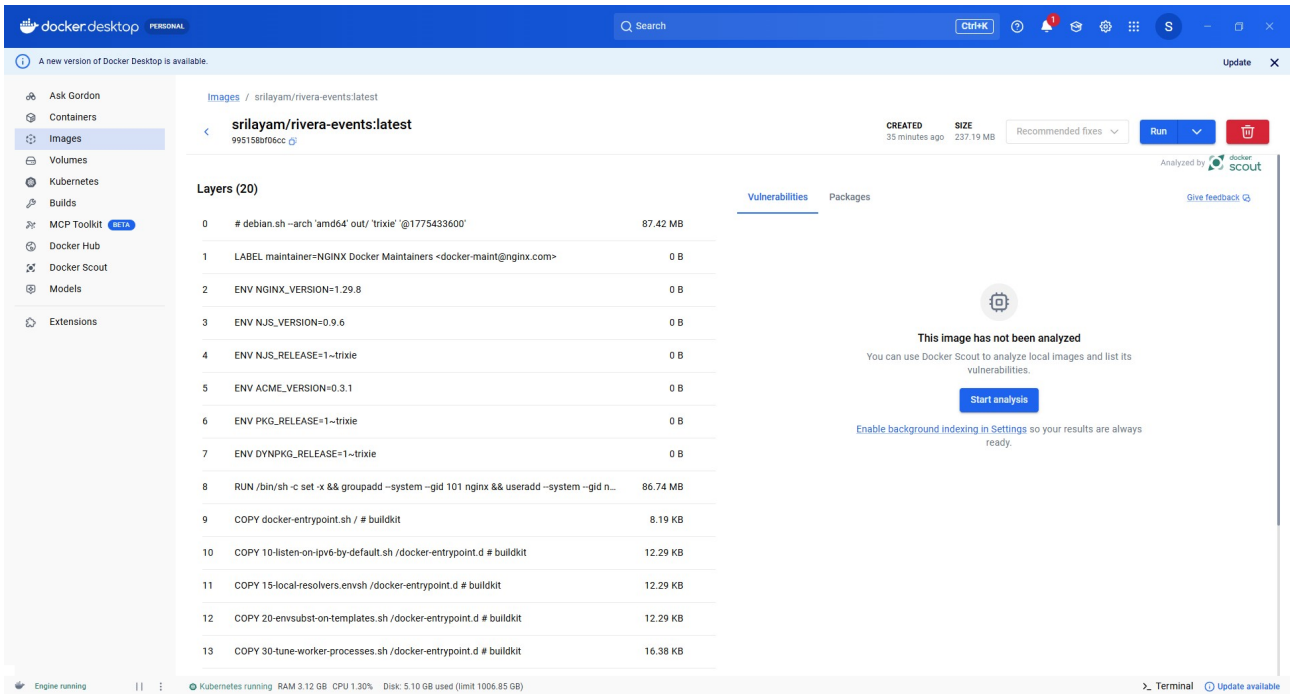
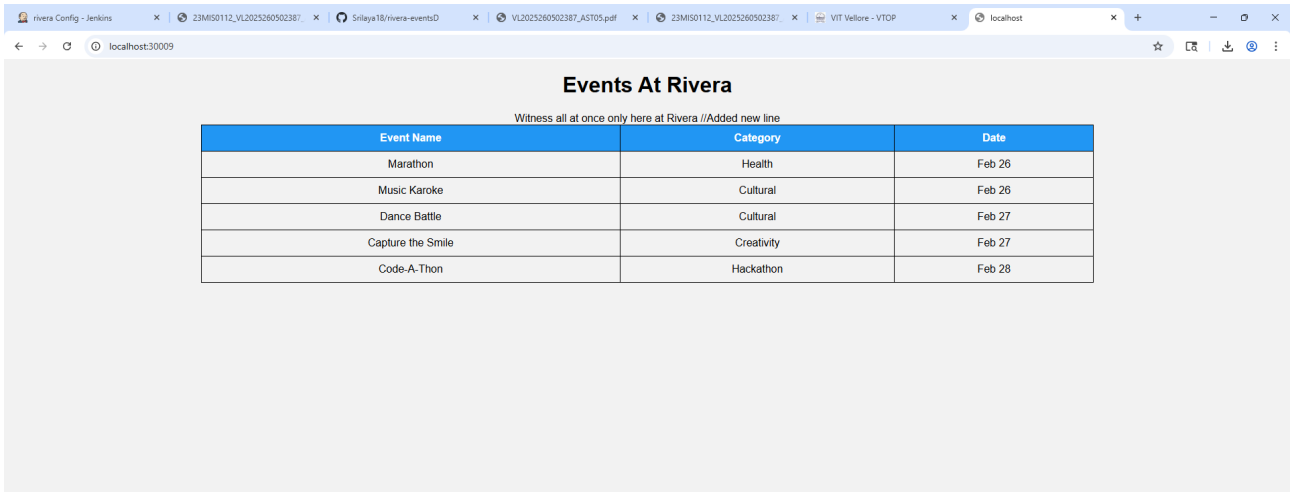


Image is visible in kuberentes





S



Git Repo: https://github.com/Srilaya18/rivera-eventsD